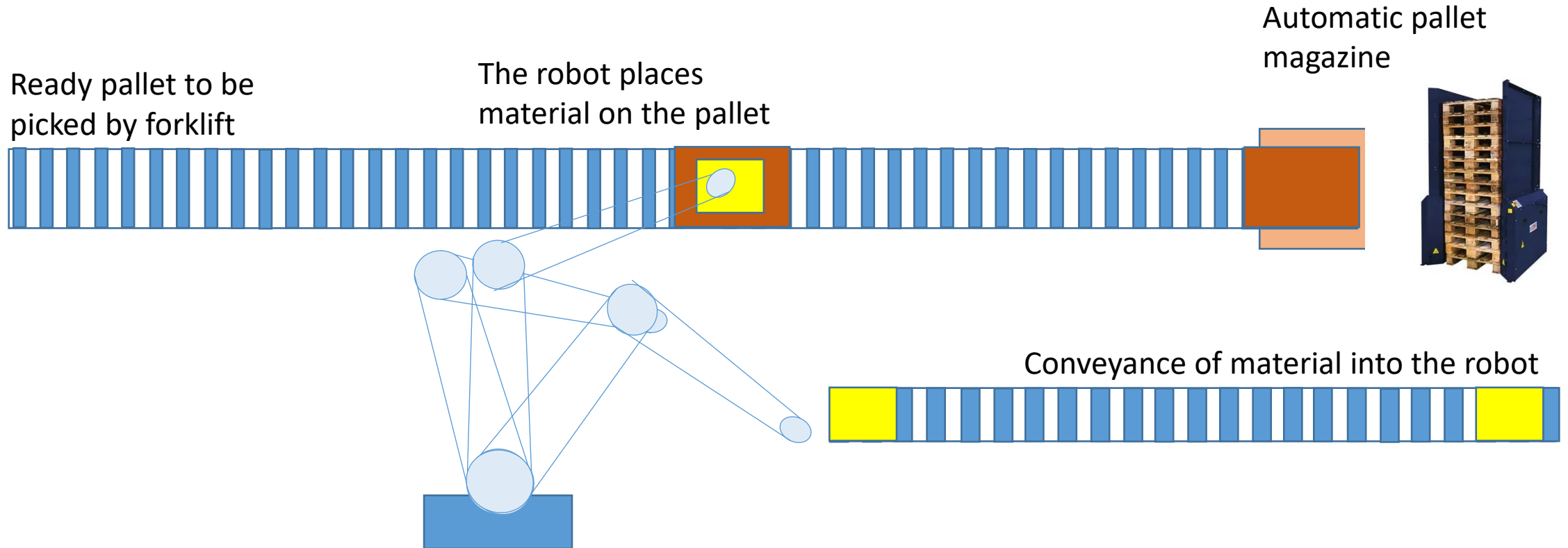


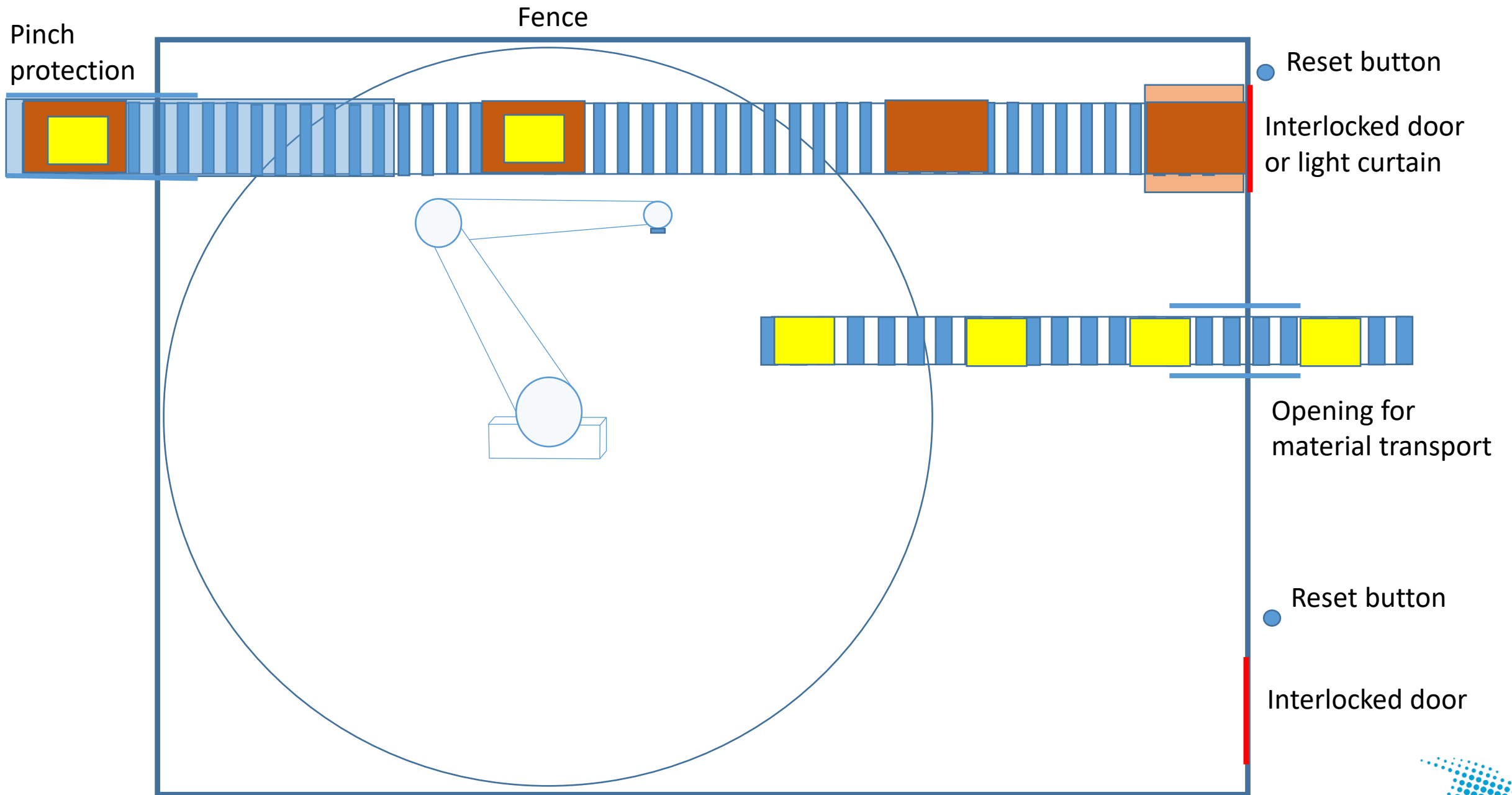
Machine Safety and Risk Reduction

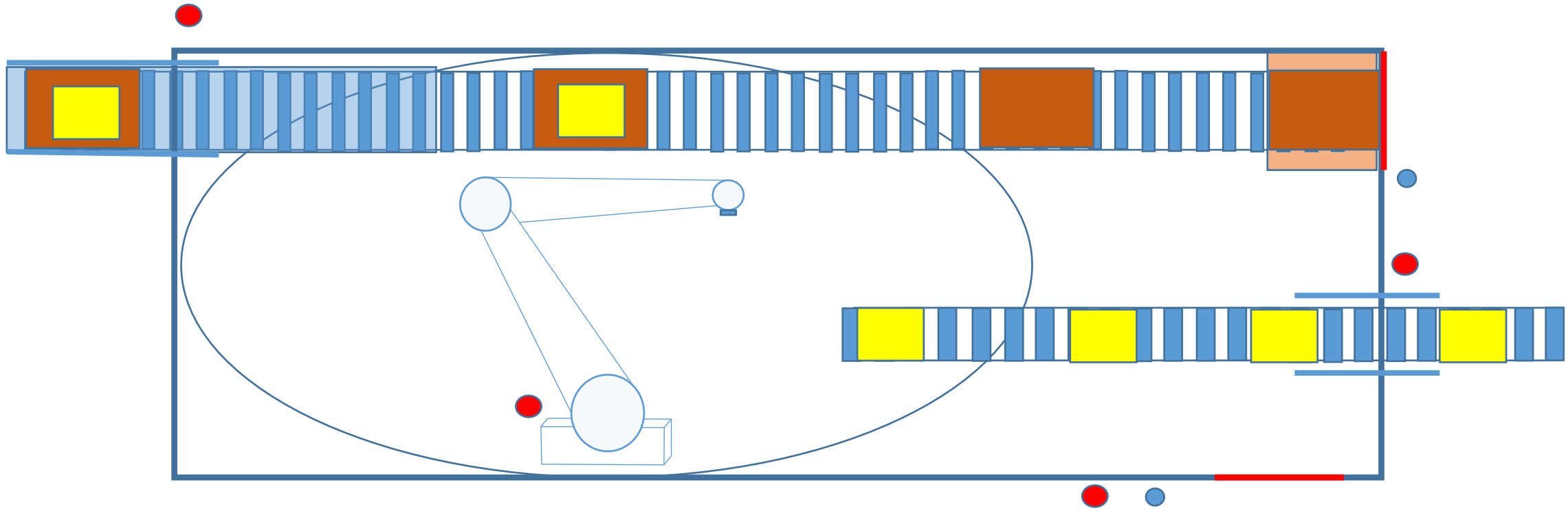
- Introduction
- Regulations and standards (Machine directive and the new regulation)
- Risk Reduction
- Performance Level (PL)
- Safety Integrity Level (SIL)
- Design categories
- Safety systems
- Safe field buses
- Safety PLC

Introduction:

An example of a palletizing robot station



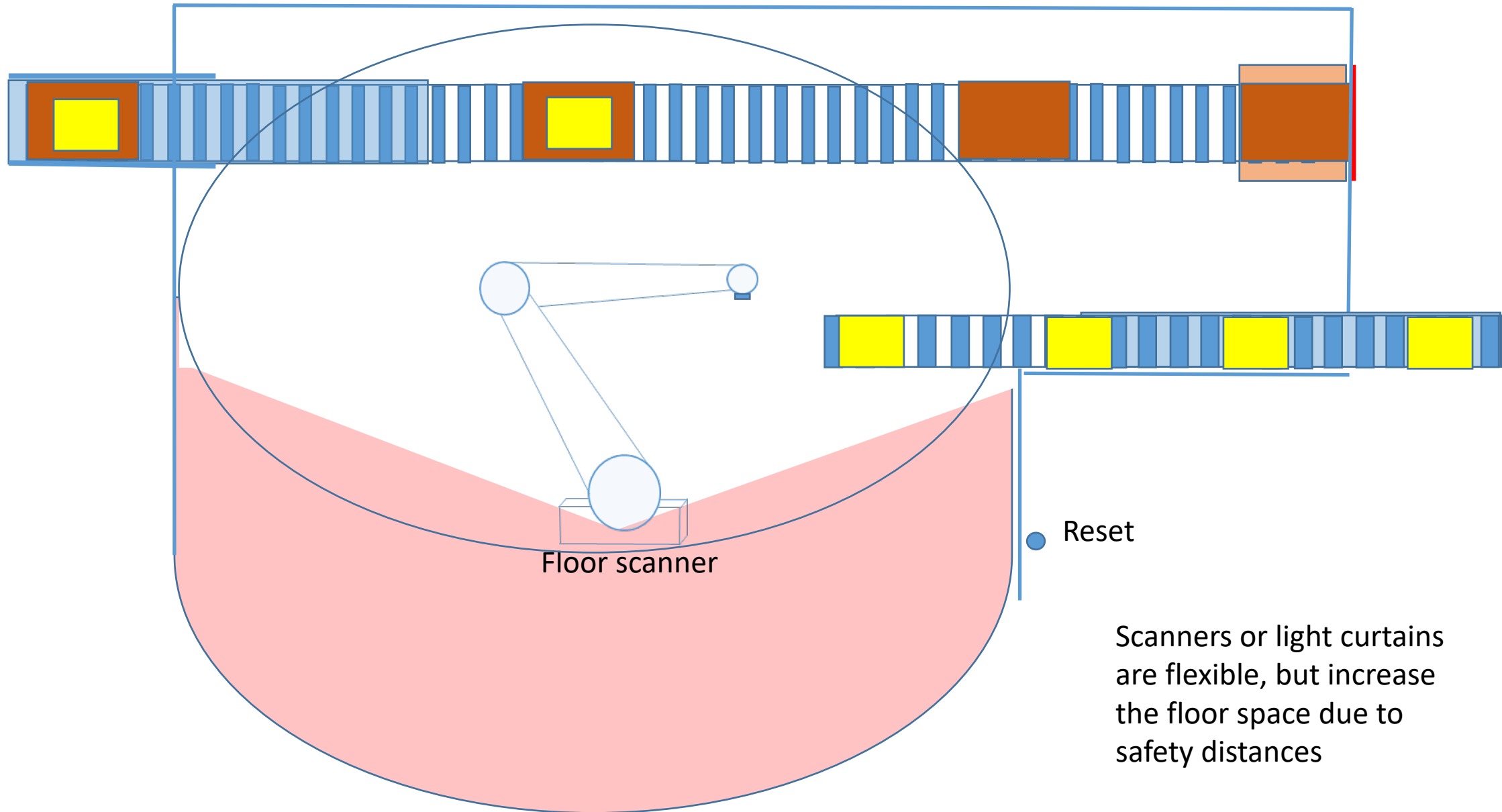




Fences can reduce the floor space area

But the robot's working area must be limited mechanically or electronically

NOTE: All emergency stops must stop all machines in the cell.



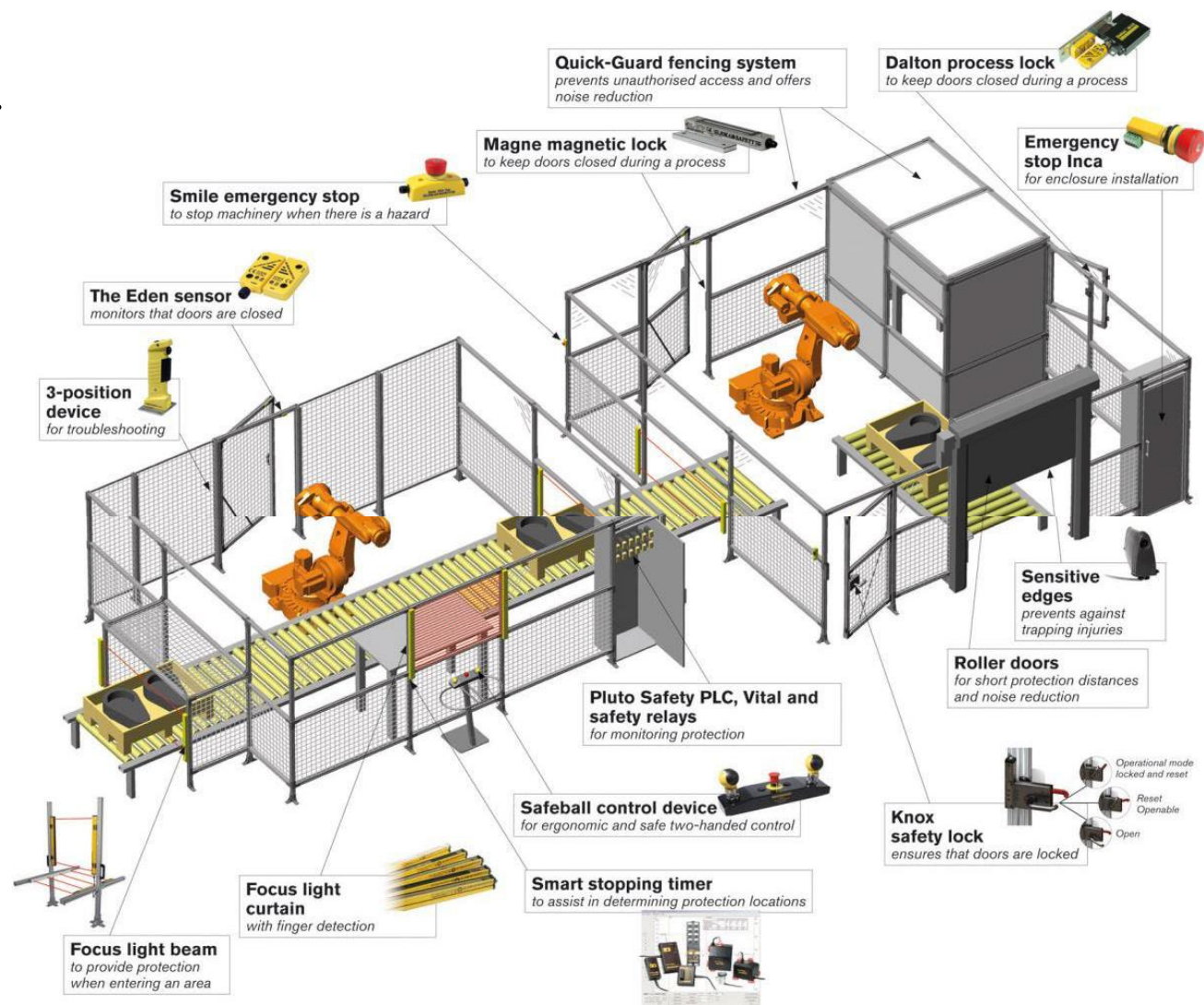
Scanners or light curtains
are flexible, but increase
the floor space due to
safety distances

History of machine safety

- 70s: New safety solutions for safety devices (e.g., Two-Hand devices) and safety control systems for presses were developed and introduced on both old and new machines.
- 80s: In the mid-eighties, the standards committee for safety in Industrial Robot Systems started. Three-position enabling devices were proposed.
- 90s: Increased work with the European standard for safety.
 - Machinery Directive (1995) updated 2006 - (2006/42/EG)
 - Machine regulation 2027
- 2000s International standards within ISO



Do we need all this?



Directives, Regulations and Standards

Regulation (EU)

2023/1230 of the European Parliament and of the Council of 14 June 2023

on Machinery
Law in all EU-countries

January 20, 2027

Examples of standards

EN ISO 12100

EN ISO 13857
EN 349
EN ISO 13849-1
EN ISO 13855

EN ISO 13850
EN 1088
EN 60204-1

EN ISO 10218-1
EN 692
EN 693

The Machinery Directive

A-standard

B1-standard
B2-standard

C-standard

2006/42/EC

AFS2008:3 makes the Machine Directive a law in Sweden

Giving basic concepts, principles for design, and general aspects that can be applied to all machinery

B1: Standards on particular safety aspects (e.g. safety distances, surface temperature, noise)

B2: Standards on safeguards, e.g. two-hand controls, interlocking devices, pressure sensitive devices, guards

Dealing with detailed safety requirements for a particular machine or group of machines

Upcoming **regulations** that will be directly applicable in all EU countries

- Machinery
- Batteries
- Cyber Resilience Act
- Cyber Security Act
- Network and Information Security
- Radio Equipment Directive
- Eco Design for Sustainable Products
- Digital Product Passport
- AI Act

European Climate Law: This regulation enshrines the EU's commitment to achieving climate neutrality by 2050.

Definition of Machine

According to the EU directive/regulation on machinery

- A group of connected components/parts with one or more moving parts (**not** powered by humans or animals) including drive system, control system, etc., which are joined together for a specific application

Not only machines included

- Related Products, including:
- **Interchangeable equipment** – Devices that modify the function of a machine.
- **Safety components** – Components that serve to ensure safety and are independently placed on the market.
- **Lifting accessories** – Equipment used for attaching loads to lifting machinery.
- **Chains, ropes, and webbing** – Specifically designed for lifting purposes.
- **Removable mechanical transmission devices** – Devices that transmit power between machines.
- **Partly Completed Machinery** – Assemblies that are almost complete machinery but cannot perform a specific application on their own and are intended to be incorporated into other machinery

Exclusions from the EU directive/regulation on machinery



- Medicine Technical products
- Transport Vehicles: Cars, Vessels, Airplanes, etc.
- Tractors and vehicles for use in farming
 - Removable and exchangeable machines connected to the tractor
- Military Equipment
- Machines or related products that are specially designed and manufactured for research purposes for temporary use in laboratories

News in the Machinery Regulations

- More detailed information than in the directive
- AI and cybersecurity are now included
- Digital user manual (Paper manual is not needed)
 - Declaration of vibration values
 - Emission and filtration systems
 - 3D printers (additive manufacturing) are included even for medical use
 - New demands on lifting equipment for persons
 - Extended requirements for partially completed machines (e.g., Robots)
 - Requirements for significant changes are defined
 - Appendix 1 – has a new structure

Three kinds of Safety in the Machinery Regulations

- Machine safety
- Cyber safety
- Process safety

Regulation of AI and Cybersecurity

- Must be included in the manufacturer's risk assessment
- Control systems
 - The machine must not perform actions outside its task and scope
 - Parameters must not be changed uncontrollably, causing risk
 - It must be possible to correct the machine
- Interface between human and machine
 - What the machine will do next
- Tracking log – information for the machine and changes to it

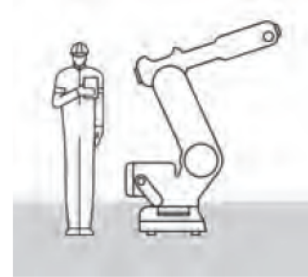
"Old" machines

Machinery that is placed on the market or put into service before 1995 in the EEA.



"New" machines

1. Machinery that is placed on the market or put into service from 1995 in the EEA.
2. All machinery that are imported to the EEA irrespective of date of origin.



Regulation (EU) 2023/1230 of the European Parliament and of the Council of 14 June 2023 on Machinery

Entered into force on 19 July 2023 and shall apply from 20 January 2027.

Stored products are possible to sell after 2027

Use of work equipment
2009/104/EC

Possible national legislation
on specific machines

Use of work equipment
2009/104/EC

Note!

Not Annex 1 - instead use
applicable directives.

The Machinery Directive
98/37/EC (1995 - 2009)
2006/42/EC (from 2010)

CE-marking + Declaration
of conformity

EMC Directive
2014/30/EU

Low Voltage Directive
2014/35/EU

Possible more directives

Figure from ABB Jokab Safety

Key Points of EN/ISO 12100:

Terminology and Principles: It defines basic terminology and principles for achieving safety in machinery design

- **Risk Assessment:** The standard outlines a systematic approach to identifying hazards and evaluating risks throughout the machine's lifecycle
- **Risk Reduction:** It provides strategies for eliminating hazards or reducing risks to acceptable levels
- **Documentation:** Guidance is given on documenting and verifying the risk assessment and risk reduction process

Continuous work environment activities EN/ISO 12100

Responsibility
Employers
Employees

**Documentation is
demanded and
must be stored for
at least 10 years**

Verifying
New risks?

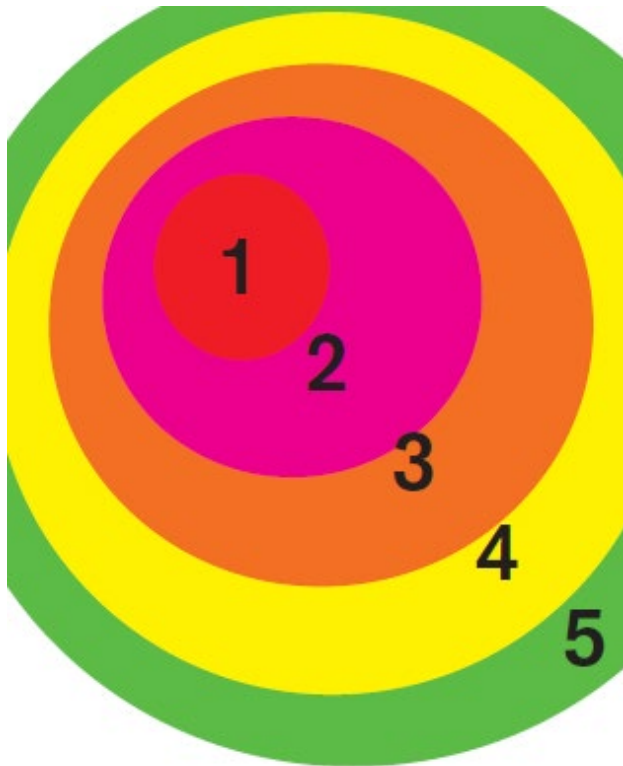


Analysis
Risk identification

Risk assessment
Planning of
Risk reduction

Application of
the risk
reduction

How to prioritize in the design phase



1. Make the machine safe by design and construction
2. Move the work tasks outside the risk area
3. Use guard/safety devices (Risk reduction)
4. Safe working routines/information
5. Warning: signs, lights, and loud

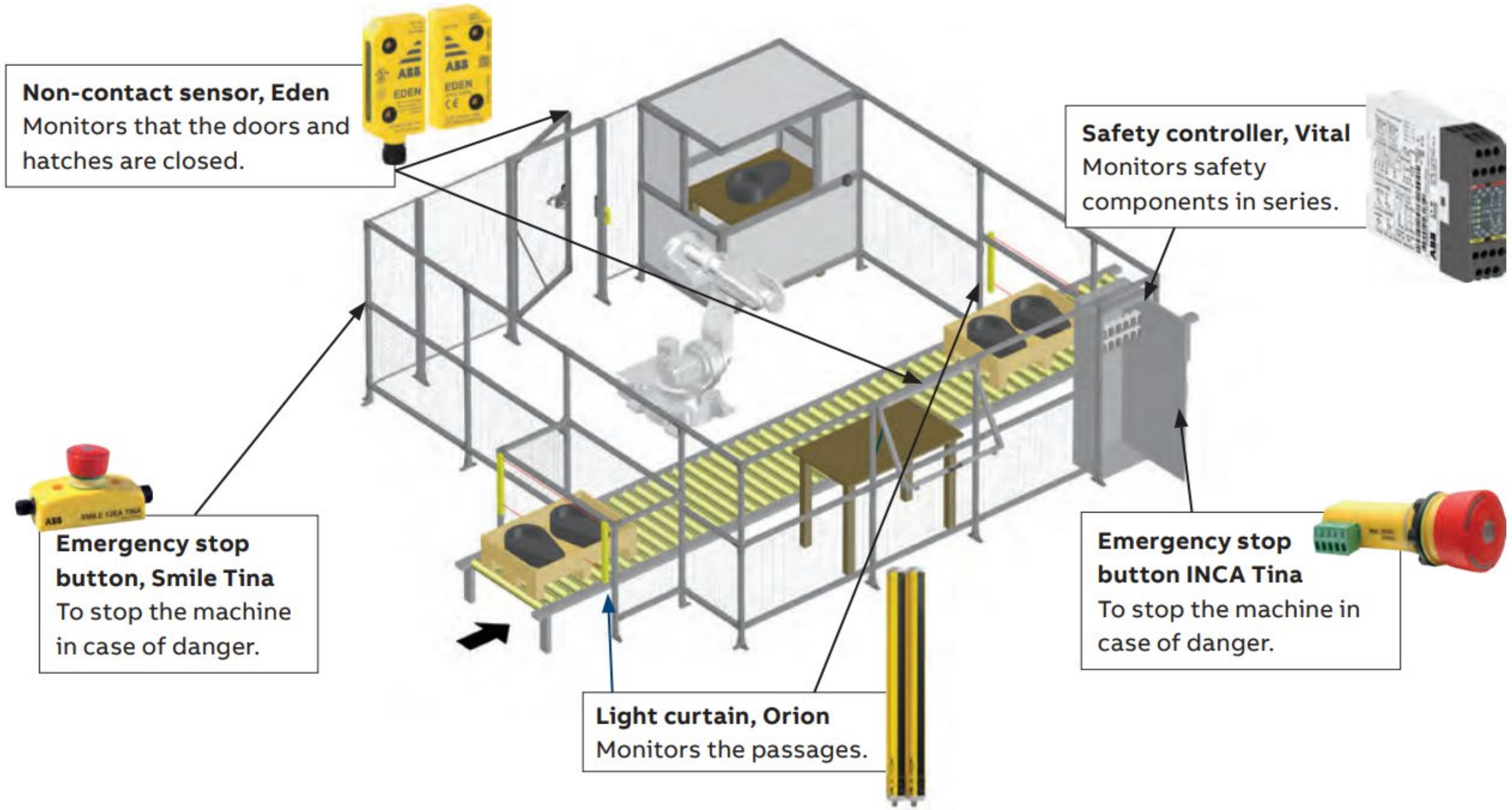
Figure from Abb Jocab Safety

Fences, light curtains, Move the work tasks outside the risk area



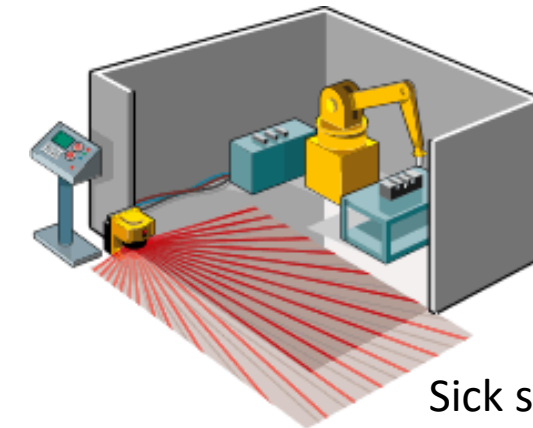
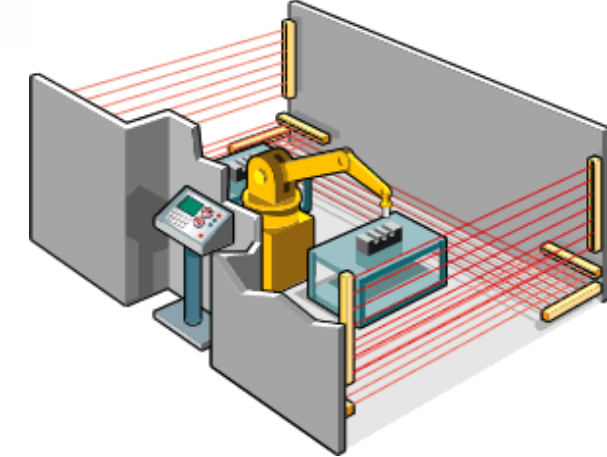
Risk reduction systems





Risk reductions

- ***Fence and gates***
 - **Advantages:** Physical stop gives a safe feeling
 - **Disadvantages:** Inflexible
- ***Light beams***
 - **Advantages:** Flexible
 - **Disadvantages:** Risk of accidentally shout downs
 - **Long safety distance** =
(Body Speed x System Reaction Time) + extra safety
- **Scanners**
 - **Advantages:** Protects restricted areas
 - **Long safety distance**



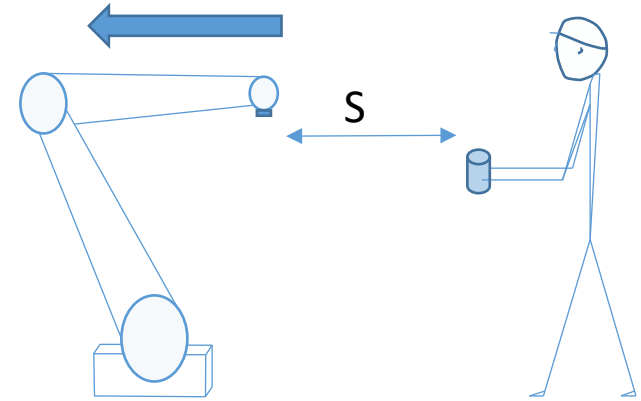
Sick safety

Safety distance

Safety of machinery EN ISO 13855

Robot standard EN ISO 10218-2

- **S Minimum separation distance**
- K_H Body or hand speed (1.6m/s – 2.0m/s)
- T_R Reaction time robot
- T_B Break time robot
- **K_R Robot speed**
- B Break distance robot
- C Safety margin due to resolution of sensors etc..



$$S = K_H(T_R + T_B) + K_R T_R + B + C$$

$$3 = 2 \quad (0.1 + 1) \quad + 1 * 0.1 + 0.5 + 0.2$$

What is a risk?

- **S**everity - Degree of injury
- **F**requency of exposure to the hazard
- **P**ossibility to detect and avoid the hazard



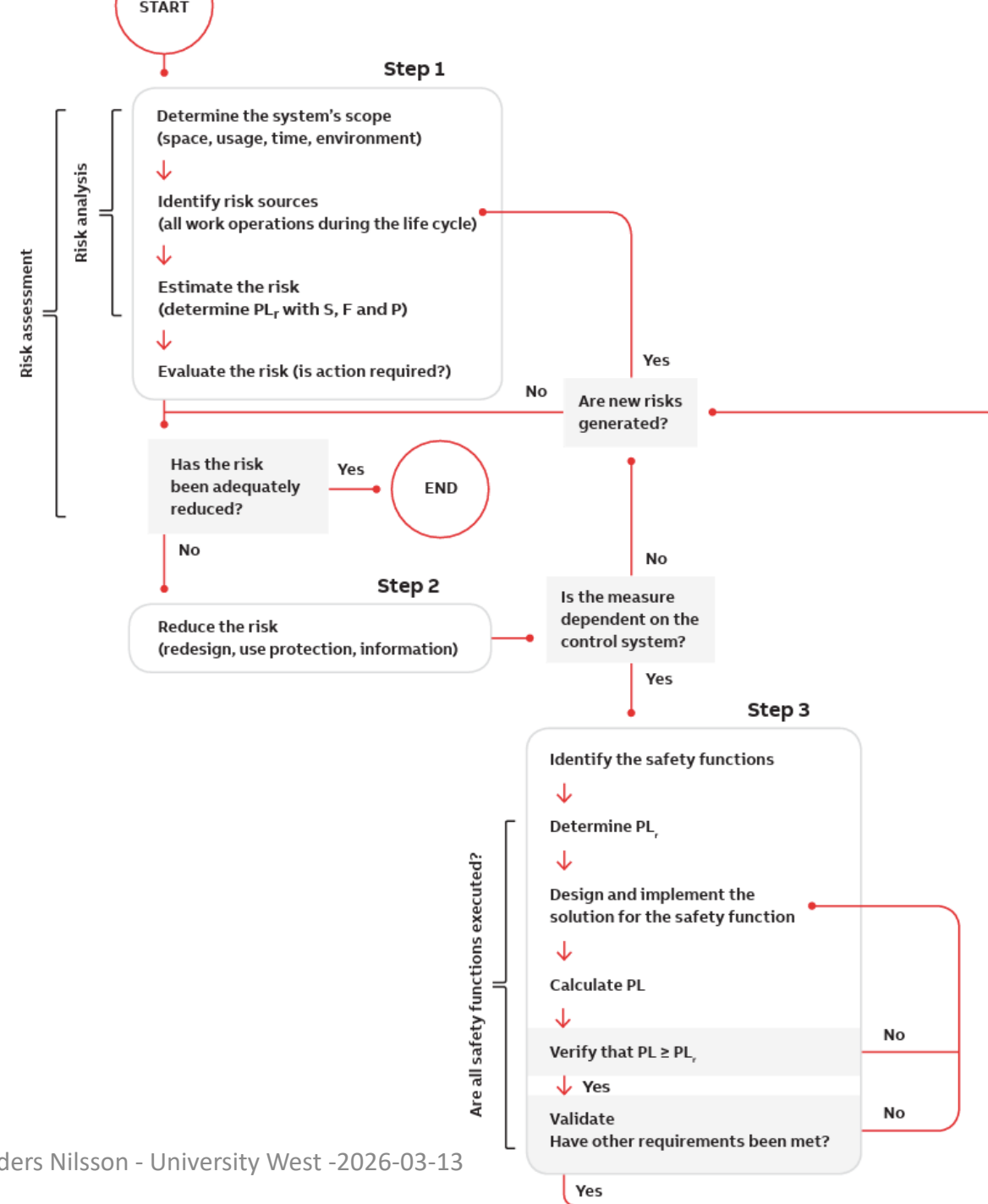
Risk reduction two parallel standards

- EN ISO 13849-1 Performance Level (PL)
 - Applicable on all risk reduction systems,
 - programmable, electronic, electric, pneumatic, and mechanical
- EN 62061 SIL
 - Simpler to use but only on electronic and programmable systems
 - Just a European standard

Definitions EN ISO 13849-1 PL

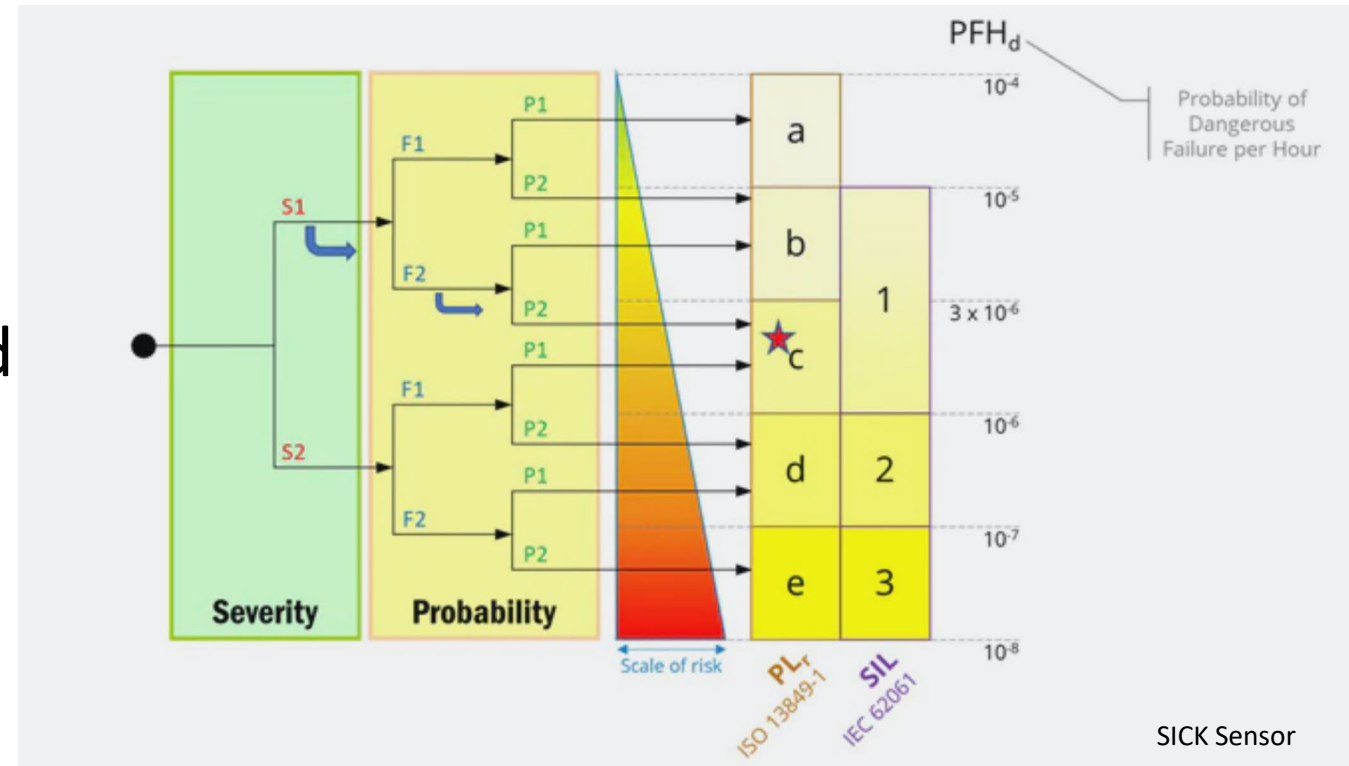
- Category Design category from B up to categories 1 to 4
- PL Performance Level
- MTTFd Mean Time To Dangerous Failure
- PFHd Probably of dangerous Failure per Hour
- B10d Average cycles until 10% of the components are ware out
- DC Diagnostic Coverage, Error detection ability
- CCF Common Cause Failure. Table of actions to protect against failure on the two channels at the same time
- TM Mission Time. The lifetime of the whole automation system

EN ISO 13849-1 PL



Judge Needed PL (a-e) and SIL (1-3)

- Degree of injury (Severity)
 - S1 Slight reversible
 - S2 Serious irreversible
- Frequency of exposure to the hazard
 - F1 Seldom and short time
 - F2 Frequent or long time
- Possibility to avoid the hazard
 - P1 Possible
 - P2 Barely possible



PL - Performance Level
SIL - Safety Integrity Level

Design categories EN ISO 13849-1 PL (Short version)

B – Single-channel sheep components

1 – Single-channel industrial components

2 – Double-channels with test function

3 – Double-channels that are supervised, faults are detected after changing the status

4 – Double-channels that are continuously supervised

Definition of design category EN ISO 13849-1 PL

Category	Summary of requirements	System behavior
B	Safety-related parts of control systems and their protective equipment shall be designed, constructed, selected, assembled, and combined in accordance with relevant standards so that they can withstand the expected influence.	The occurrence of a fault can lead to the loss of the safety function.
1	Requirements of B shall apply. Well-tries components and well-tries safety principles shall be used.*	The occurrence of a fault can lead to the loss of the safety function, but the probability of occurrence is lower than for category B.
2	Requirements of B and the use of well-tries safety principles shall apply. The safety function shall be checked at suitable intervals by the machine control system.	The occurrence of a fault can lead to the loss of the safety function between the checks. The loss of the safety function is detected by the check.
3	Requirements of B and the use of well-tries safety principles shall apply. Safety-related parts shall be designed so that - a single fault in any of these parts does not lead to the loss of the safety function and - whenever reasonably practicable, the single fault is detected.	When a single fault occurs, the safety function is always performed. Some, but not all, faults will be detected. Accumulation of undetected faults can lead to the loss of the safety function.
4	Requirements of B and the use of well-tries safety principles shall apply. Safety-related parts shall be designed so that - a single fault in any of these parts does not lead to the loss of the safety function and - single faults are detected at or before the next demand upon the safety function, but that if this detection is not possible, an accumulation of undetected faults shall not lead to the loss of the safety function.	When faults occur, the safety function is always performed. The faults will be detected in time to prevent the loss of the safety function.

How to calculate the degree of the design category?

Mean Time To Dangerous Failure

EN ISO 13849-1 PL

$$MTTF_d = \frac{B_{10d}}{0.1 * \text{Cycles per year}}$$

B10d Average cycles until 10% of the components are ware out

Level of MTTFd	Range
1 Low	3 years ≤ MTTFd < 10 years
2 Medium	10 years ≤ MTTFd < 30 years
3 High	30 years ≤ MTTFd < 100 years

Diagnostic Coverage (DC)

EN ISO 13849-1

DU = Dangerous undetected failures

DD = Dangerous detected failures

D = Total failures = DU + DD

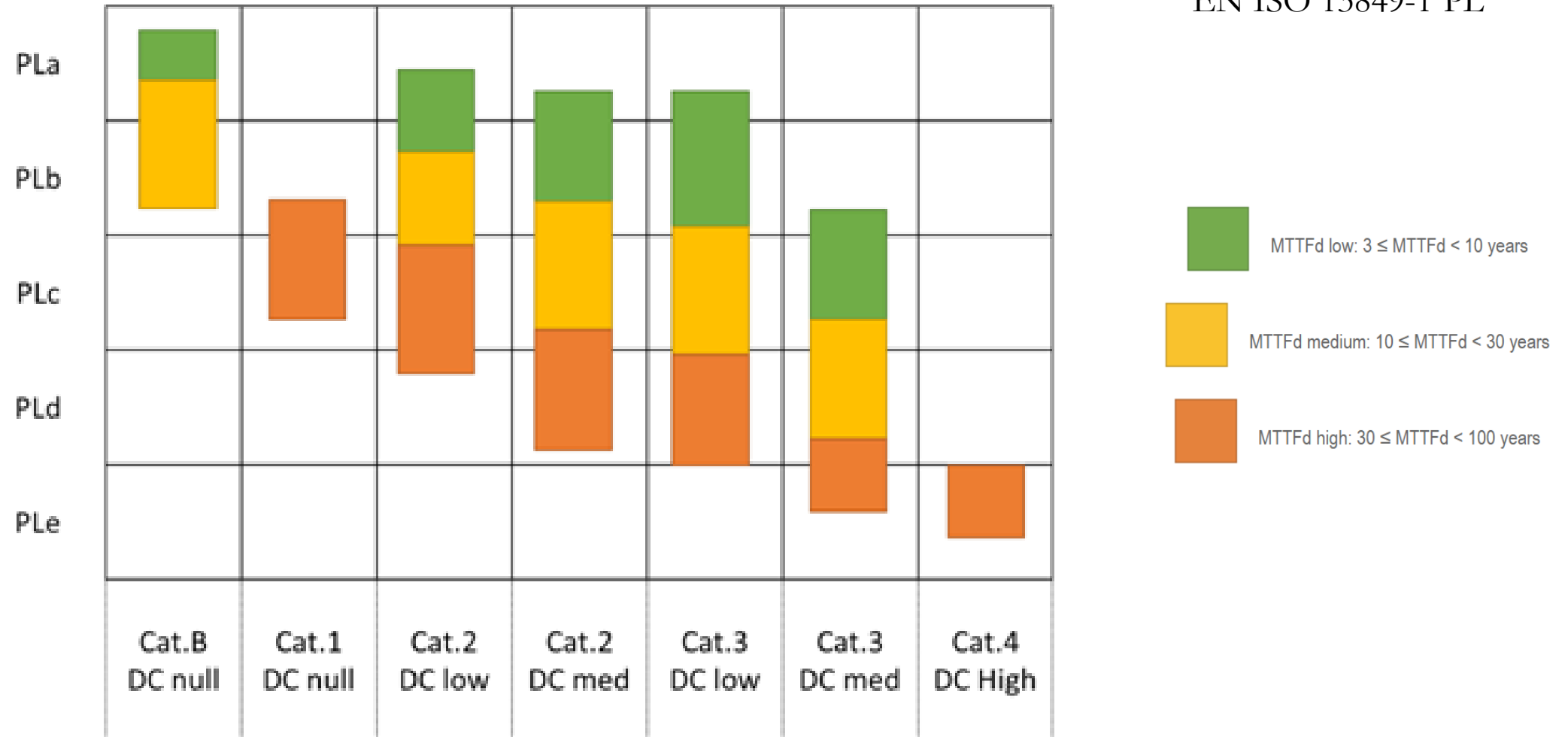
$$DC = DD / D * 100$$

How easy is a failure to be discovered (FMEA IEC60812)

DC average	Range
None	DC < 60%
Low	60 % ≤ DC < 90 %
Medium	90 % ≤ DC < 99 %
High	99 % ≤ DC < 100 %

Determination of Design Category

EN ISO 13849-1 PL



<https://www.sensy.com/en/technology/standards/iso-13849-safety-of-machinery>

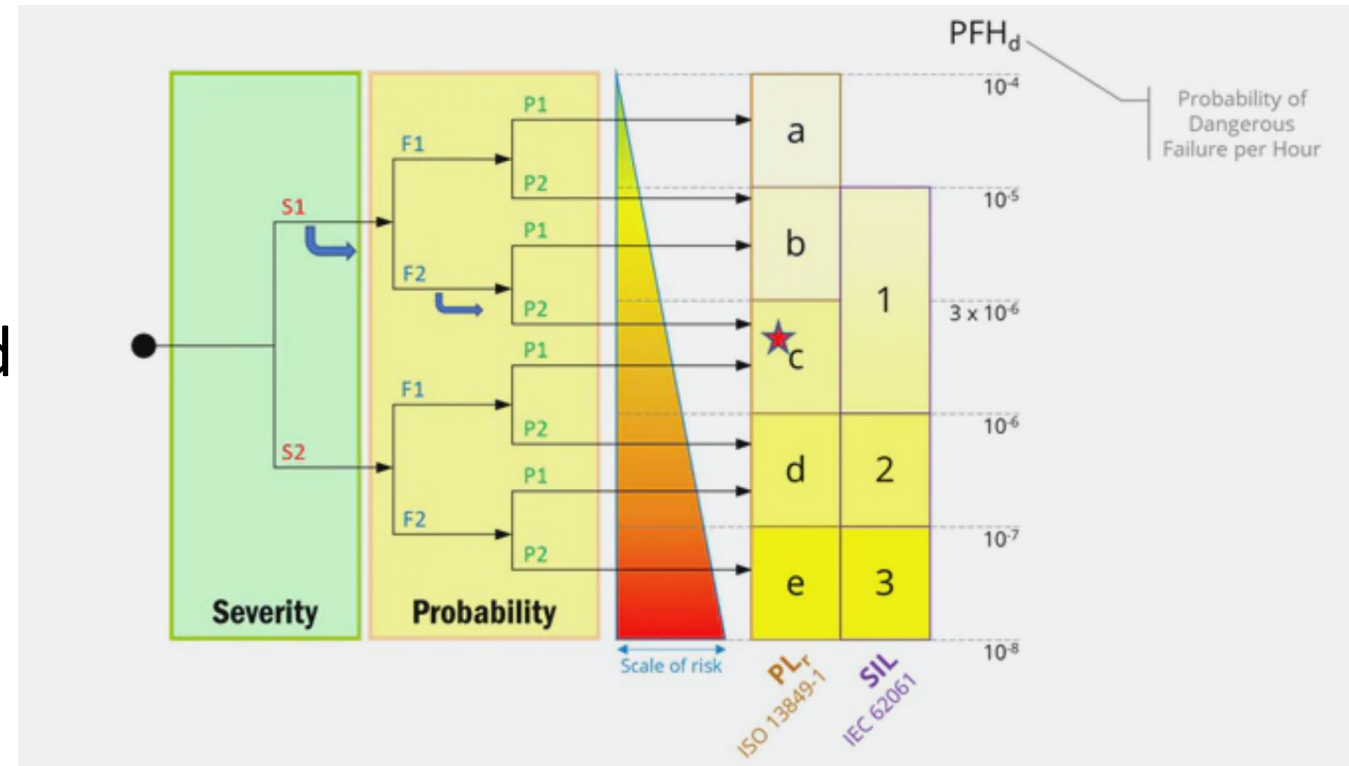
EN 62061 SIL (Europe only)

- SIL Safety Integrity Level
- SILCL SIL Claim Limit
- PFH_d Probably of dangerous failure per hour
- T₁ Lifetime

SIL	PFH _d
1	$\geq 10^{-6}$ till $< 10^{-5}$
2	$\geq 10^{-7}$ till $< 10^{-6}$
3	$\geq 10^{-8}$ till $< 10^{-7}$

Judge Needed PL (a-e) and SIL (1-3)

- Degree of injury (Severity)
 - S1 Slight reversible
 - S2 Serious irreversible
- Frequency of exposure to the hazard
 - F1 Seldom and short time
 - F2 Frequent or long time
- Possibility to avoid the hazard
 - P1 Possible
 - P2 Barely possible



EN 62061 SIL assessment

Company namn	Risk assessment and safety measures	Document No.: _____
Product: _____		Part of: _____
Issued by: _____		☐ Pre risk assessment
Date: _____	Black area = Safety measures required Grey area = Safety measures recommended	☐ Intermediate risk assessment
		☐ Follow up risk assessment

Consequences	Severity Se	Class Cl					Frequency and duration, Fr		Probability of hzd. event, Pr		Avoidance Av
		3 - 4	5 - 7	8 - 10	11 - 13	14 - 15					
Death, loosing an eye or arm	4	SIL 2	SIL 2	SIL 2	SIL 3	SIL 3	<= 1hour	5	Very high	5	
Permanent, loosing fingers	3		OM	SIL 1	SIL 2	SIL 3	> 1 hr - <=day	5	Likely	4	
Reversible, medical attention	2			OM	SIL 1	SIL 2	>1day - <=2wks	4	Possible	3	Impossible
Reversible, first aid	1				OM	SIL 1	>2wks - <=year	3	Rarely	2	Possible
							> 1year	2	Negligible	1	Likely

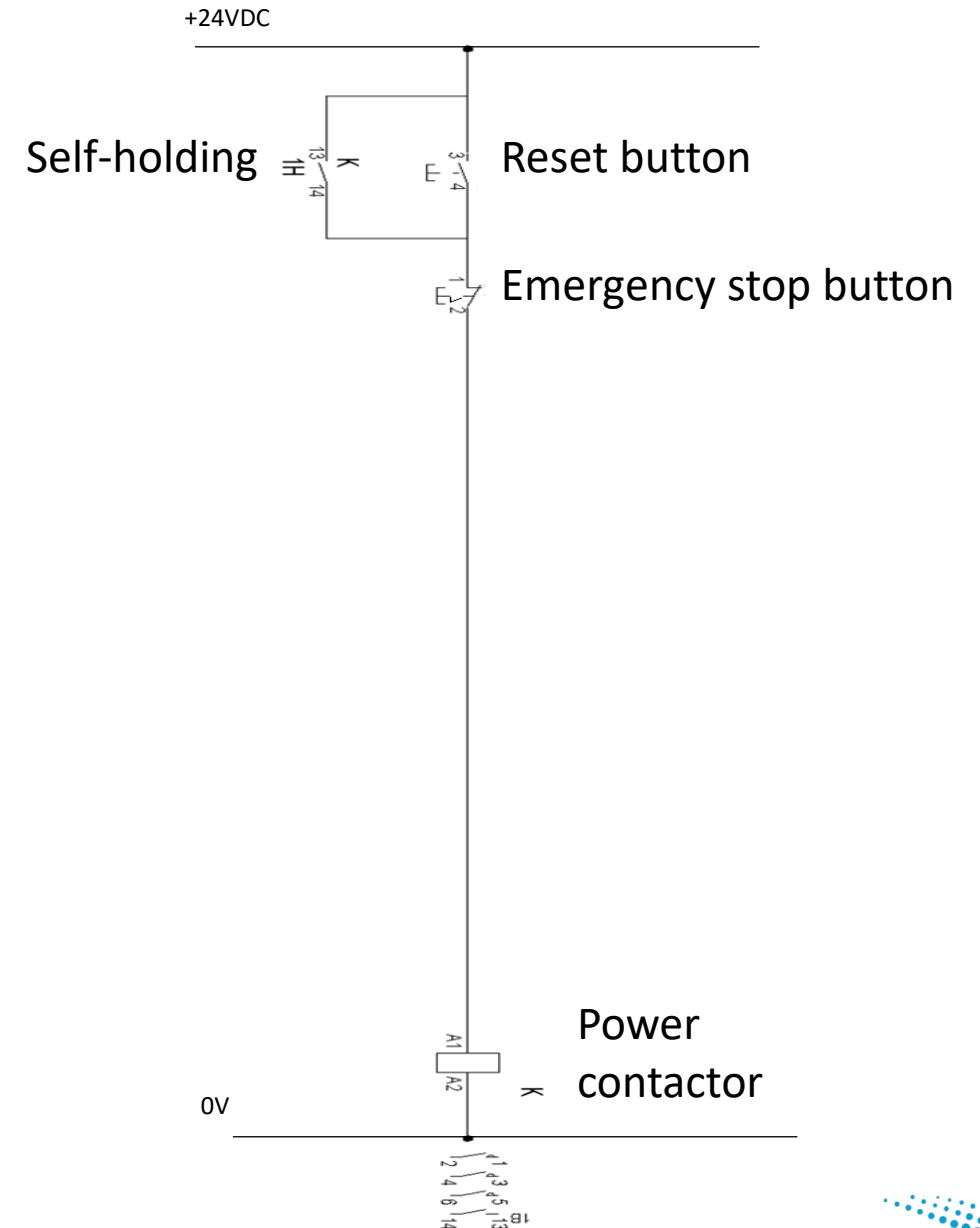
Ser. No.	Hzd. No.	Hazard	Se	Fr	Pr	Av	Cl	Safety measure	Safe
		Cutting risk magazine	3	4	3	5	12	Openable cover with interlock	

Comments

Example of Design categories

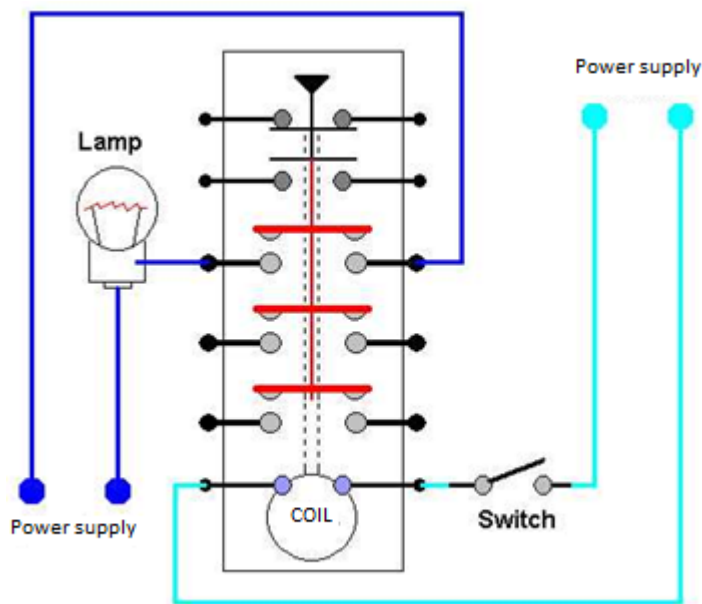
Category B (Basic) to Category 1

- Industrial standard components for (Category 1)
- Single channel
- Self-holding, protects against immediate start after power failure according to the machine directive

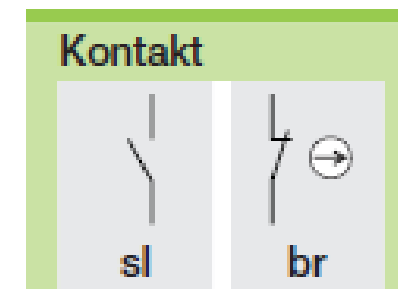


Mechanical interlocked contacts

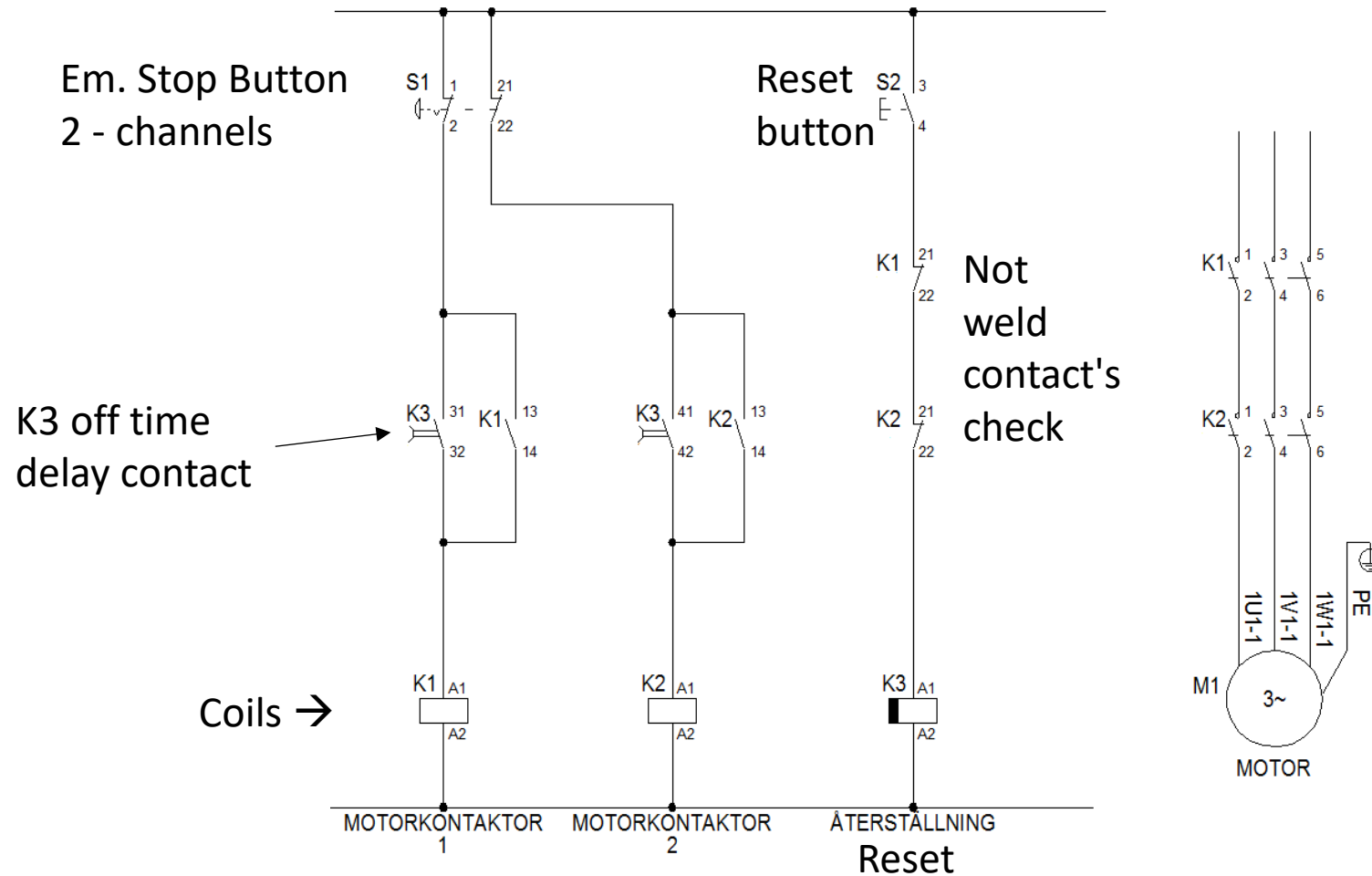
Safe relays, contactors, and approximately switches



- Weld no transfer
- Force to break contacts

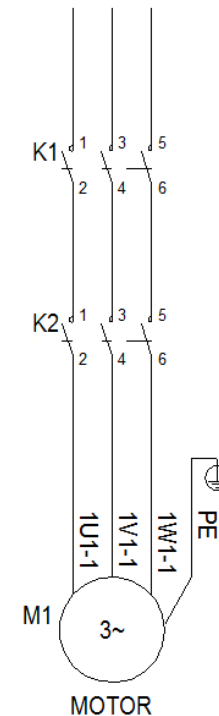


Contactor-based solution by weld no transfer contacts (Well-tried principles)



S1 Emergency stop
S2 Reset button
K1 Motor Contactor
K2 Motor contactor
M1 3-Phase Motor

The contact 21-22 on K1
and K2 supervise that K1
and K2 have released
mechanically to reset by
the reset button S2



Electronic Safety “relay” double and supervision Category 3



ABB JOKAB Safety RT9

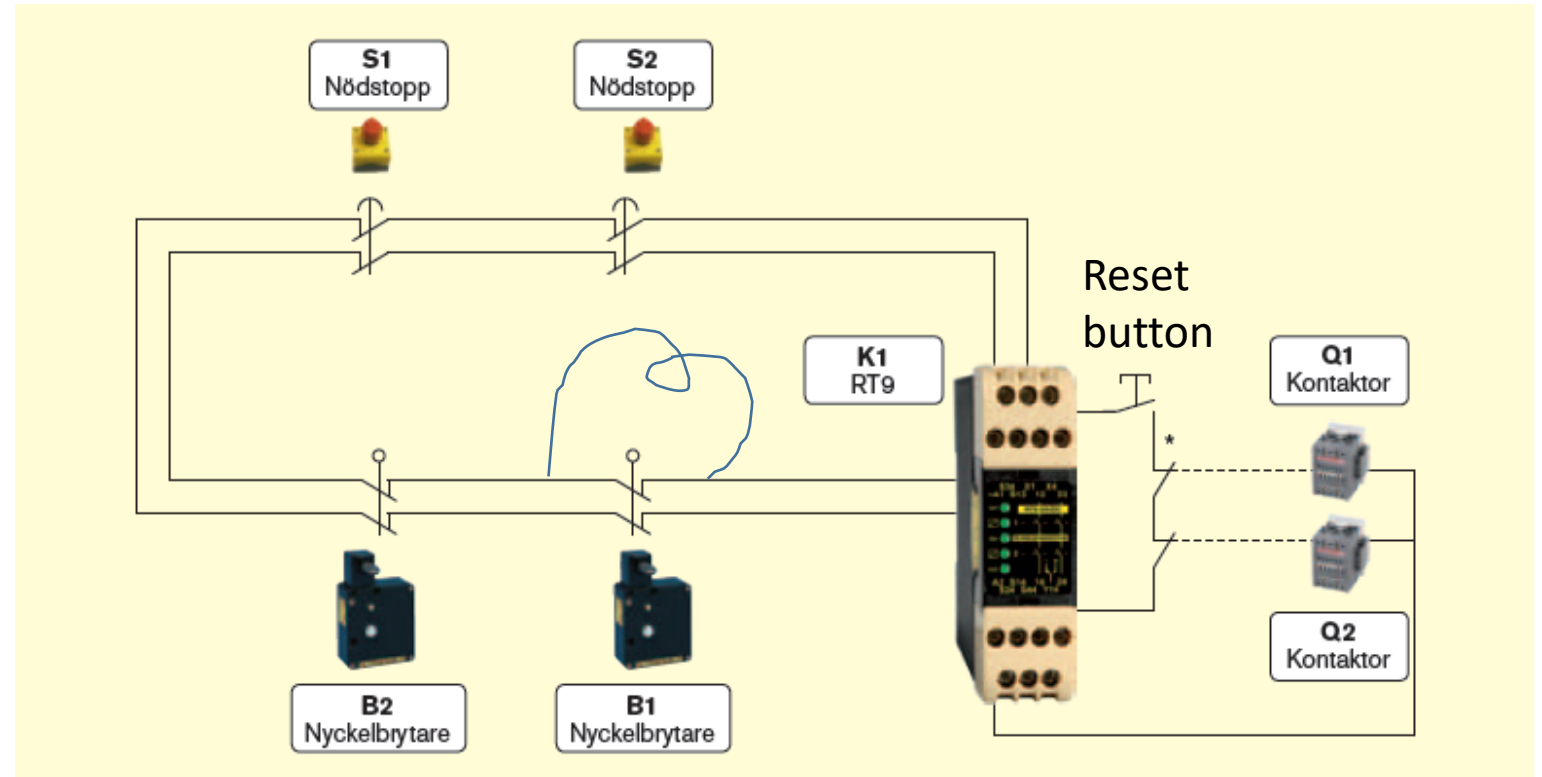
Category 4/PL e

(EN ISO 13849-1:2008)

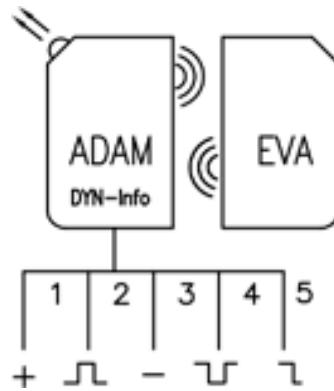
SIL 3 (EN 62061:2005)

PFHd = 9.55E-09

$B_{10d} = 10^7$ cykler

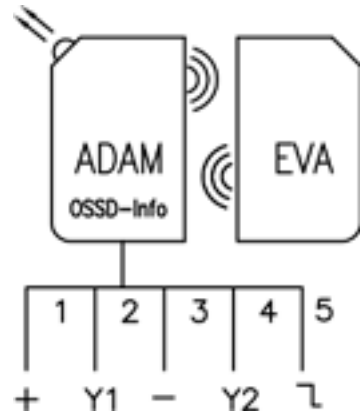


Active dynamic sensors and buttons (Cat4)

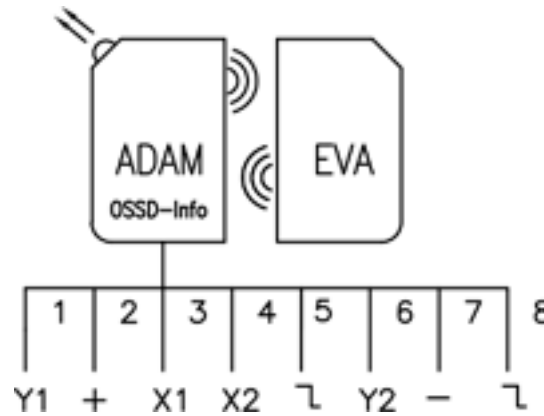


- Pulsed signal instead of two channels
- The “ADAM” module inverts the signal. The safety system can then detect a shortcut

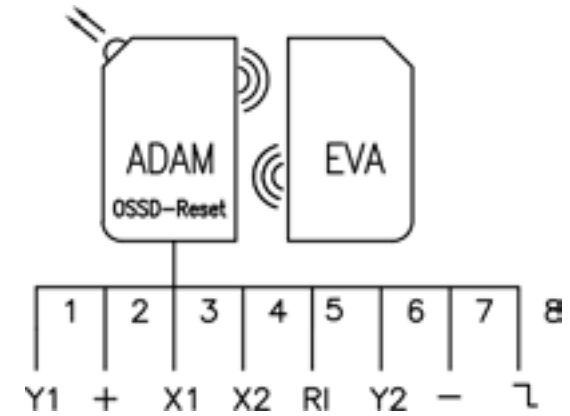
OSSD Output Signal Switching Device (Cat4)



- 1) +24V
- 2) OSSD 1 Ut
- 3) 0V
- 4) OSSD 2 Ut
- 5) Information

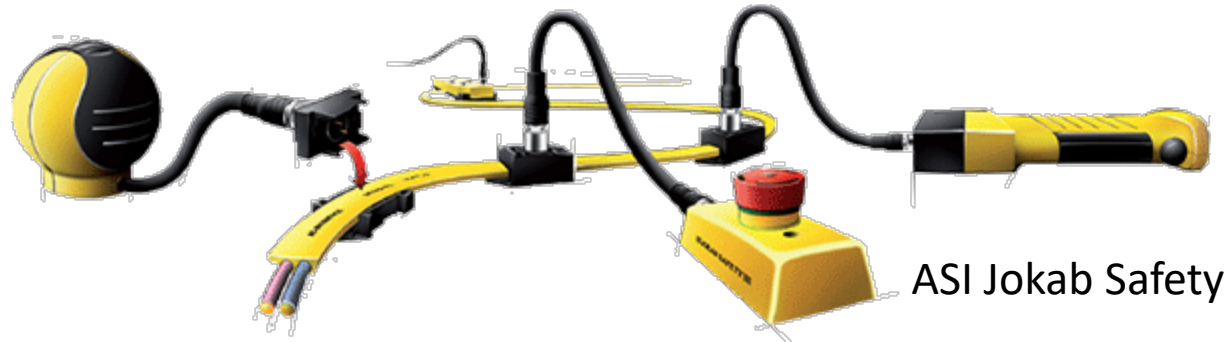


- 1) OSSD 1 Ut
- 2) +24V
- 3) OSSD 1 In
- 4) OSSD 2 In
- 5) Information
- 6) OSSD 2 Ut
- 7) 0V
- 8) Information



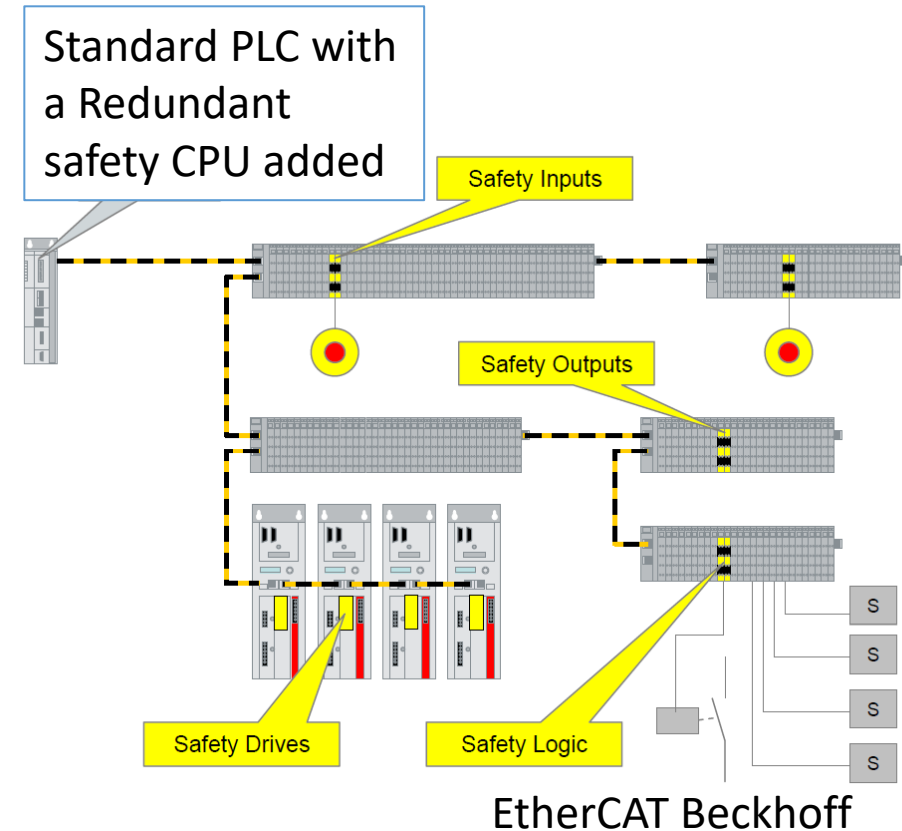
- 1) OSSD 1 Ut
- 2) +24V
- 3) OSSD 1 In
- 4) OSSD 2 In
- 5) Reset/Indikering
- 6) OSSD 2 Ut
- 7) 0V
- 8) Information

Safe Fieldbus communication and safe PLC

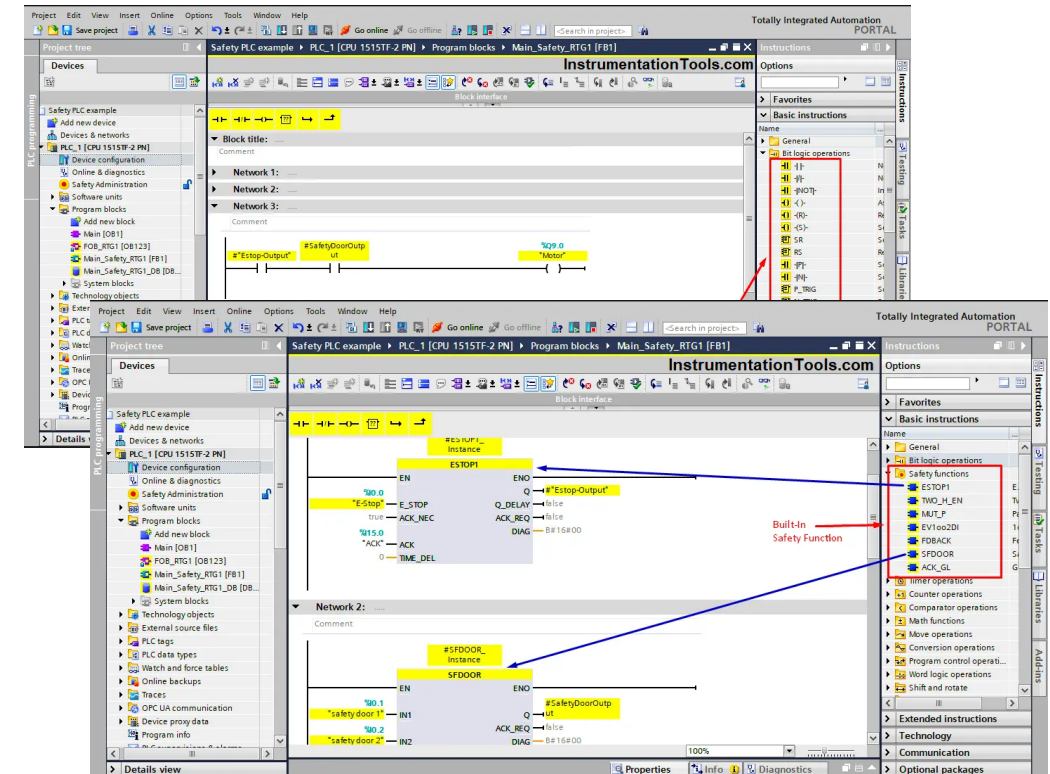
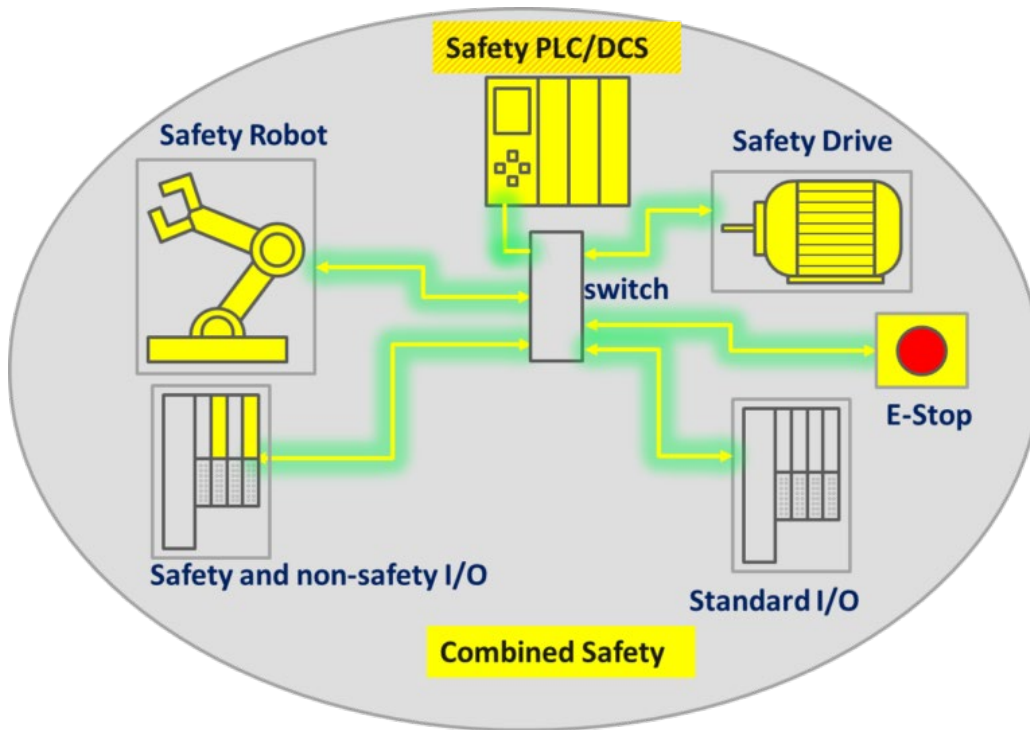


Common safety buses

- ProfiNet Safe
- EtherCAT Safe
- ASI Safe
- Safety Bus-P (Piltz)
- Pluto Safety Bus (ABB JOKAB Safety)



Programmable Safety Logic (Safety PLC)



<https://instrumentationtools.com/safety-plc-coding-practices/>

<https://profinetuniversity.com/functional-safety/safety-basics-conventional-vs-combined-safety/>

Safety PLC EN ISO 13849-1 Apendix J

- The program in a Safety PLC must be
 - Simple as possible
 - Readable
 - Understandable
 - Use the included validated function blocks as much as possible
- Validation
- Function test
- Documentation

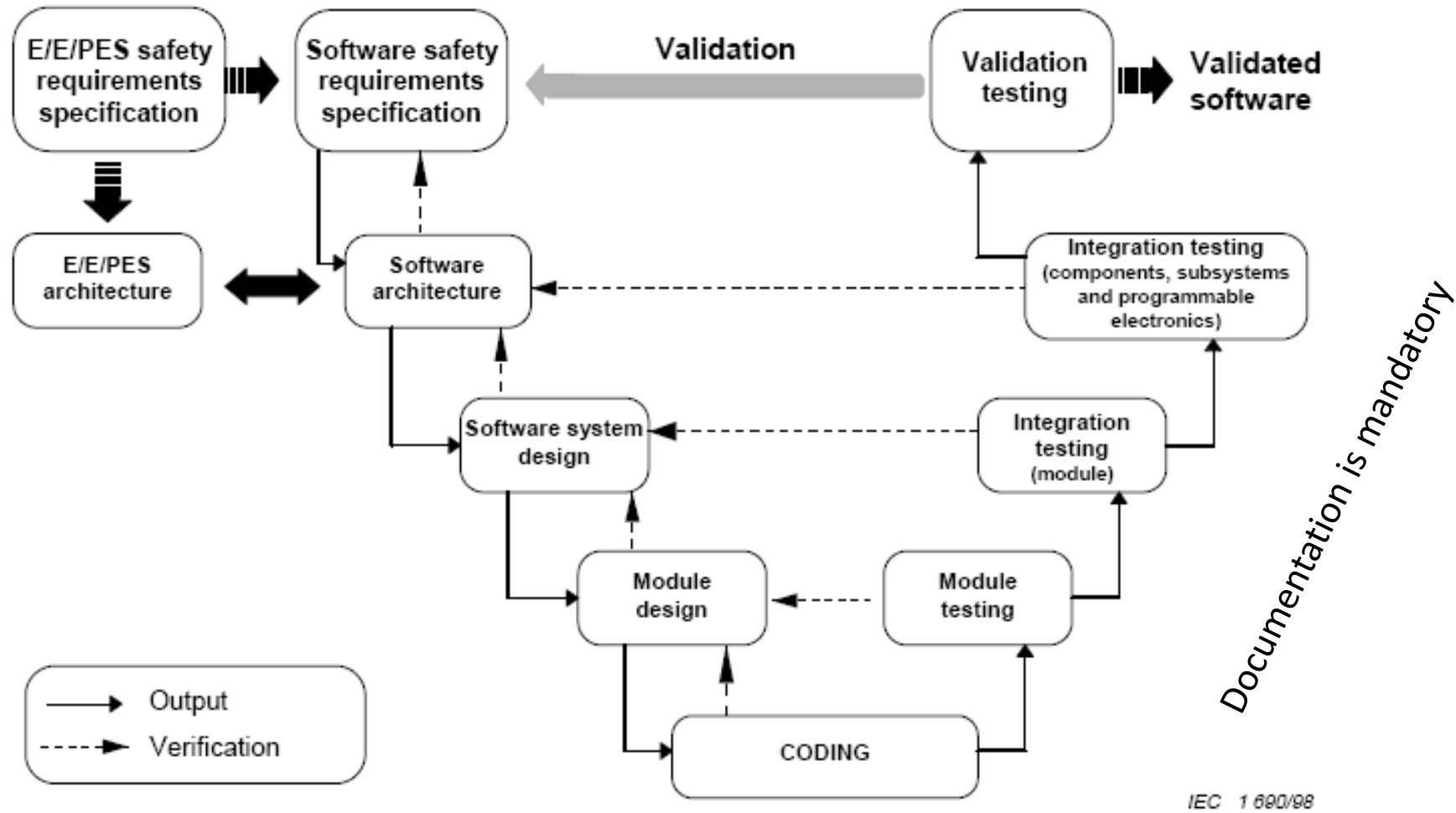


Figure 5 – Software safety integrity and the development lifecycle (the V-model)